

WASP-104b

R-2615

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_160103 · created 2026-08-28T22:48:21+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457390.894281 +/- 3.6e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.013
Transit depth (Rp/Rs)^2	1.96 +/- 0.36 [%]
Orbital Inclination (inc)	83.53 +/- 1.32
Airmass coefficient 1 (a1)	1.02 +/- 0.012
Airmass coefficient 2 (a2)	-0.0146 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#3 - [334, 249]
Optimal Aperture	2.93
Optimal Annulus	12.98
Transit Duration (day)	0.073 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.013 vs 0.1214 (deviation 1.43σ)
Duration fit vs published	0.073 d vs 0.0742 d (deviation 0.1σ)
Mid-time O–C	-0.3 min
Depth SNR	10.8
Residual scatter	1.23 %

Light curve

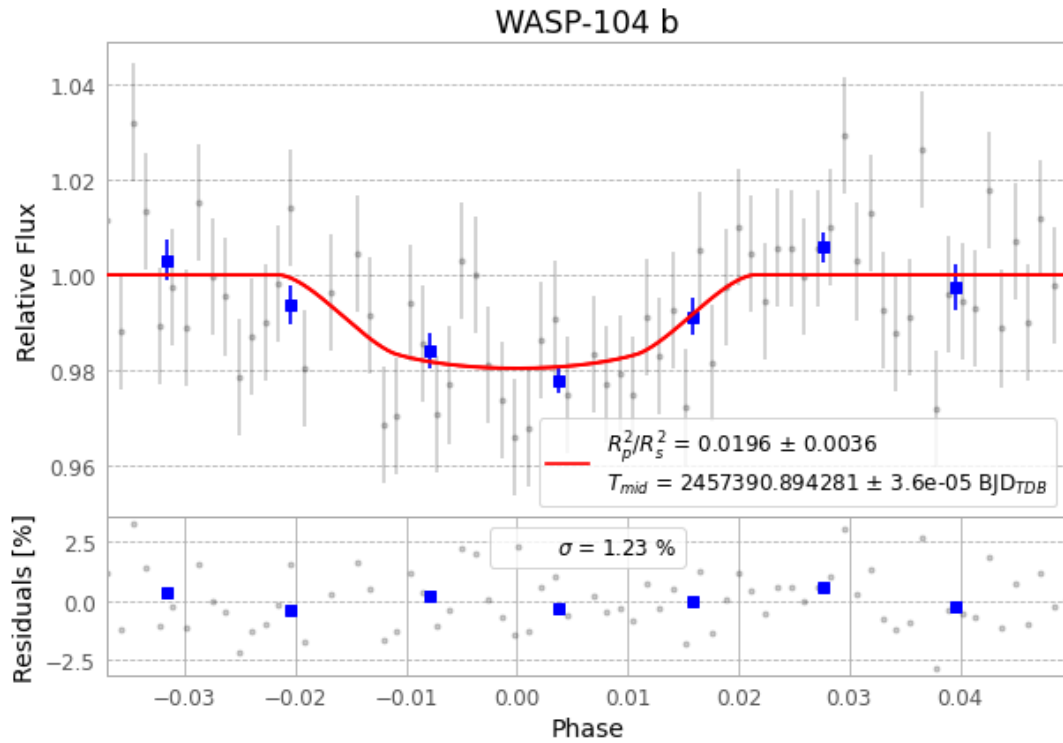


Figure 1 — FinalLightCurve_WASP-104 b_2016-01-03.png (SHA-256 4d178c8b9fc2c7bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 057f5c60ae7b52ad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), WASP-104160103074812.FITS → WASP-104160103112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160103.log (7f03757329c5...), FinalLightCurve_WASP-104 b_2016-01-03.png (4d178c8b9fc2...), FinalLightCurve_WASP-104 b_2016-01-03.csv (861a9fe3c0f3...)

Compute resources

Wall time 123.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 22:48Z.